

Numerical solver 2-link IK

$$J(\theta_1, \theta_2) = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix} \quad f(\theta_1, \theta_2) = \begin{bmatrix} x(\theta_1, \theta_2) \\ y(\theta_1, \theta_2) \end{bmatrix}$$

$$x = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2)$$

$$\frac{\partial x}{\partial \theta_1} = -L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2), \quad \frac{\partial x}{\partial \theta_2} = -L_2 \sin(\theta_1 + \theta_2)$$

$$\frac{\partial y}{\partial \theta_1} = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2), \quad \frac{\partial y}{\partial \theta_2} = L_2 \cos(\theta_1 + \theta_2)$$

Idea behind numerical solver:

$$\Delta p = \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \quad \Delta p = J \Delta \theta, \text{ maps joint change to end-effector change.}$$

$$\Delta \theta = \begin{bmatrix} \Delta \theta_1 \\ \Delta \theta_2 \end{bmatrix} \quad \text{error} = p_{\text{target}} - p_{\text{current}}$$

Since we know p_{target} , J and p_{current}

$$J \Delta \theta \approx \text{error}$$

$\Delta \theta \approx J^+ \text{error}$. we use the pseudoinverse

(Moore-Penrose inverse) to solve.